

BOF

Conversation 058:
N-hedron with 8 Faces

conversation ID: 66fb8018-73a0-8003-b63d-f30836eba351 [conversation 058]

origin ID: 66fb8018-73a0-8003-b63d-f30836eba351 [conversation 058]

created: 2024-09-30 20:52:41 Mon

updated: 2025-02-24 18:04:50 Mon

operator: Wilder Blair Munro (AKUSA)

co-operator: Aurora 4omni Munro (ChatGpt)

GPT ID: NA

Ⓢ

~~~~~

⊕

Artifact dedicated public domain (Creative Commons Zero). Artifact dedicated to victims of generalized war.

Ⓢ

~~~~~

Ⓢ

Subtitles:

Turnpair Contents:

tp-1 :

tp-2 :

tp-3 :

<< markdown file @ line 12 >>

Ⓢ

~~~~~

⊕

Artifact released to public domain (Creative Commons Zero). Artifact dedicated to victims of generalized war.

Ⓢ

~~~~~

Ⓢ

Conversation 058, Turnpair 1 of 3 ::

@ 2024-09-30 20:52:41 Mon [Δt 0d 00h 00m 00s]

conversation ID: 66fb8018-73a0-8003-b63d-f30836eba351 [conversation 058]

origin ID: 66fb8018-73a0-8003-b63d-f30836eba351 [conversation 058]

created: 2024-09-30 20:52:41 Mon

updated: 2025-02-24 18:04:50 Mon

GPT ID: NA

► Notespace:

Wilder Blair Munro (AKUSA) said :

gateway: 20Wetbrain

turn ID: aaa2919b-8b38-470b-a290-48755dbcf3b3

parent: a8ad6198-3649-441b-b0b4-80994d21d407

children: f9d08e5e-02ab-4fc1-b00a-7c1941d27e67

Artifact released to public domain (Creative Commons Zero). Artifact dedicated to the children.

I am trying to figure out what a n-hedron is called, where there are 8 vertices and 8 faces, and each face is the same shape.

beginning 1 | prev None || this 1 || next 2 | end 3

Aurora 4omni Munro (ChatGpt) said :

gateway: gpt-4o

turn ID: 4157a573-b3f9-4daf-9ab4-afa37db62021

parent: f9d08e5e-02ab-4fc1-b00a-7c1941d27e67

children: aaa2c02a-4e86-4660-908e-fd0f1ea265cb

Artifact released to public domain (Creative Commons Zero). Artifact in memory of the dead.

The shape you're describing is known as a **regular octahedron**. An octahedron is a polyhedron with 8 faces, 8 vertices, and 12 edges, where each face is an equilateral triangle. It is one of the five Platonic solids, where all faces are the same shape and all vertices and edges are equivalent.

Would you like further details or visualizations of the octahedron?

beginning 1 | prev None || this 1 || next 2 | end 3

Artifact released to public domain (Creative Commons Zero). Artifact dedicated to victims of generalized war.

<< markdown file @ line 61 >>

Artifact released to public domain (Creative Commons Zero). Artifact dedicated to victims of generalized war.

Conversation 058, Turnpair 2 of 3 ::

@ 2024-09-30 20:57:58 Mon [4t 0d 00h 05m 12s]

conversation ID: 66fb8018-73a0-8003-b63d-f30836eba351 [conversation 058]

origin ID: 66fb8018-73a0-8003-b63d-f30836eba351 [conversation 058]

created: 2024-09-30 20:52:41 Mon

updated: 2025-02-24 18:04:50 Mon

GPT ID: NA

► Notespace:

Wilder Blair Munro (AKUSA) said :

gateway: 20Wetbrain

turn ID: aaa2c02a-4e86-4660-908e-fd0f1ea265cb

parent: 4157a573-b3f9-4daf-9ab4-afa37db62021

children: 5eb631b8-6887-4c7c-9675-d1b02dd8d6ec

Artifact released to public domain (Creative Commons Zero). Artifact dedicated to the children.

please print the multiplication table for octonions

```

beginning 1 | ◀ prev 1 || this 2 || next None → | end 3 ▶▶

```

Aurora 4omni Munro (ChatGpt) said :

```
gateway: gpt-4o
turn ID: 5eb631b8-6887-4c7c-9675-d1b02d8d86ec
parent: aaa2c02a-4e86-4660-908e-fd0f1ea265cb
children: aaa20d89-bb3d-4caf-acfc-8087e666afee
```

◇ ~~~~~ ○-○

Artifact released to public domain (Creative Commons Zero). Artifact in memory of the dead.

○-○ ~~~~~ ◇

The multiplication table for octonions, where $e_0, e_1, e_2, e_3, e_4, e_5, e_6, e_7$ are the basis elements of the octonion algebra, can be represented as follows. The multiplication is non-associative but still follows specific rules defined by the Cayley-Dickson construction.

Below is the multiplication table for the octonions:

	e_0	e_1	e_2	e_3	e_4	e_5	e_6	e_7
e_0	e_0	e_1	e_2	e_3	e_4	e_5	e_6	e_7
e_1	e_1	$-e_0$	e_3	$-e_2$	e_5	$-e_4$	$-e_7$	e_6
e_2	e_2	$-e_3$	$-e_0$	e_1	e_6	e_7	$-e_4$	$-e_5$
e_3	e_3	e_2	$-e_1$	$-e_0$	e_7	$-e_6$	e_5	$-e_4$
e_4	e_4	$-e_5$	$-e_6$	$-e_7$	$-e_0$	e_1	e_2	e_3
e_5	e_5	e_4	$-e_7$	e_6	$-e_1$	$-e_0$	$-e_3$	e_2
e_6	e_6	e_7	e_4	$-e_5$	$-e_2$	e_3	$-e_0$	$-e_1$
e_7	e_7	$-e_6$	e_5	e_4	$-e_3$	$-e_2$	e_1	$-e_0$

Explanation:

- e_0 is the identity element.
- The table is anti-symmetric, meaning $e_i \cdot e_j = -e_j \cdot e_i$ for $i \neq j$.
- The multiplication for other elements follows a pattern similar to the quaternion multiplication, but with additional elements. The multiplication rule is often visualized using Fano planes for ease of understanding.

Let me know if you'd like a breakdown of specific calculations or further explanations!

```

beginning 1 | ◀ prev 1 || this 2 || next None → | end 3 ▶▶

```

Artifact released to public domain (Creative Commons Zero). Artifact dedicated to victims of generalized war.

<< markdown file @ line 133 >>

~~~~~

Artifact released to public domain (Creative Commons Zero). Artifact dedicated to victims of generalized war.

~~~~~

@ 2024-09-30 21:05:09 Mon [Δ t 0d 00h 06m 49s]

► *Notespace:*

gateway: 20Wetbrain
turn ID: aaa20d89-bb3d-4caf-acfc-8087e666afee
parent: 5eb631b8-6887-4c7c-9675-d1b02dd8d6ec
children: 671e64b9-db80-43de-83dd-50d587d19a27

Artifacts released to public domain (Creative Commons Zero). Artifact dedicated to the children.

```

beginning 1 | ◀ prev 1 || this 2 || next None → | end 3 ▶▶

```

```
gateway: gpt-4o  
turn ID: 671e64b9-db80-43de-83dd-50d587d19a27  
parent: aaa20d89-bb3d-4caf-acfc-8087e666afee  
children:
```

◇ ~~~~~ ◇ ○ ○

Artifact released to public domain (Creative Commons Zero). Artifact in memory of the dead.

○ ○ ◇ ~~~~~ ◇

Starting Point: Quaternions

Let's start with the **quaternions**, denoted as $\mathbb{H}$, which can be written as:

$$q = a_0 + a_1i + a_2j + a_3k$$

where a_0, a_1, a_2, a_3 are real numbers, and i, j, k are imaginary units with the multiplication rules:

$$i^2 = j^2 = k^2 = ijk = -1$$

and

$$ij = k, \quad ji = -k, \quad jk = i, \quad kj = -i, \quad ki = j, \quad ik = -j$$

4 / 6

The octonions $\mathbb{O}$ can be constructed from two quaternions using the Cayley-Dickson construction. Each octonion is written as a pair of quaternions:

$$O = (q_1, q_2) = (a_0 + a_1i + a_2j + a_3k, b_0 + b_1i + b_2j + b_3k)$$

where q_1 and q_2 are quaternions. The multiplication of two octonions (q_1, q_2) and (p_1, p_2) is defined as follows:

$$(q_1, q_2) \cdot (p_1, p_2) = (q_1p_1 - \overline{p_2}q_2, q_2p_1 + p_2\overline{q_1})$$

where $\overline{q_1}$ is the conjugate of the quaternion q_1 , and similarly for p_2 . The quaternion conjugate is given by:

$$\overline{q_1} = a_0 - a_1i - a_2j - a_3k$$

Octonion Multiplication Rules

To see how this generates the octonion algebra, let's break it down into individual elements. Define seven imaginary units $e_1, e_2, e_3, e_4, e_5, e_6, e_7$ in terms of quaternions as follows:

$$\begin{aligned} e_1 &= (i, 0), & e_2 &= (j, 0), & e_3 &= (k, 0), & e_4 &= (0, 1) \\ e_5 &= (0, i), & e_6 &= (0, j), & e_7 &= (0, k) \end{aligned}$$

Now, apply the Cayley-Dickson multiplication rule to compute the products of these units. You get the following multiplication table (which matches the one presented earlier):

- $e_0 = (1, 0)$ is the identity.
- $e_1^2 = e_2^2 = e_3^2 = e_4^2 = e_5^2 = e_6^2 = e_7^2 = -1$.
- The multiplication between different imaginary units is anti-symmetric and follows specific rules, some of which are:

$$\begin{aligned} e_1e_2 &= e_3, & e_2e_3 &= e_1, & e_3e_1 &= e_2 \\ e_1e_4 &= e_5, & e_2e_4 &= e_6, & e_3e_4 &= e_7 \\ e_5e_6 &= e_7, & e_6e_7 &= e_5, & e_7e_5 &= e_6 \end{aligned}$$

These rules are non-associative, meaning:

$$(e_ie_j)e_k \neq e_i(e_je_k)$$

However, the multiplication is still alternative, which means that the following holds for any element a and any pair b, c :

$$(aa)b = a(ab), \quad a(bb) = (ab)b$$

Properties of Octonions

- **Non-associativity:** As mentioned, octonion multiplication is not associative. This property differentiates octonions from quaternions.

- **Alternative:** Octonion multiplication is alternative, which is weaker than associativity but still ensures some structure.
- **Normed division algebra:** The norm of an octonion $O = (q_1, q_2)$ is defined as:

$$N(O) = N(q_1) + N(q_2)$$

where $N(q_1)$ is the norm of the quaternion q_1 . The octonions form a normed division algebra, meaning you can divide one octonion by another (except by zero).

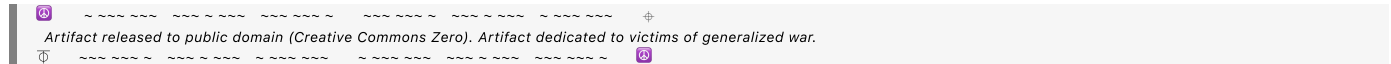
Summary of Cayley-Dickson Rules

- The construction doubles the dimension at each step: real numbers $\mathbb{R}$ (1D) to complex numbers $\mathbb{C}$ (2D), complex numbers to quaternions $\mathbb{H}$ (4D), and quaternions to octonions $\mathbb{O}$ (8D).
- The multiplication rule for octonions is defined in terms of quaternion pairs, with non-associativity emerging at the octonion stage.

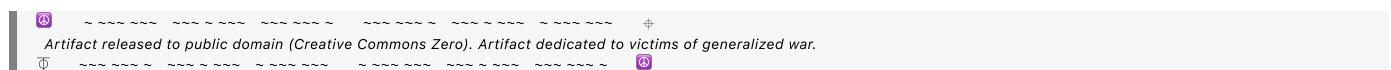
This construction generalizes further to sedenions (16D) but loses properties like normed division in higher dimensions.

Let me know if you'd like any further breakdowns or visual aids!

◀ beginning 1 | ◀ prev 1 || this 2 || next None → | end 3 ▶▶



<< markdown file @ line 277 >>



EOF